UN R62 (version 1.1.0)

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1. SCOPE

This Regulation applies to vehicles of categories L_1 to L_7, 1/ if fitted with handlebars.

1/ As defined in Annex 7 to the Consolidated Resolution on the Construction of Vehicles (R.E.3),(document TRANS/WP.29/78/Rev.1/Amend.2 as last amended by Amend.4).

2. DEFINITIONS

For the purposes of this Regulation:

2.1. "Approval of a vehicle" means the approval of a vehicle type with regard to its protection against unauthorized use;

2.2. "Vehicle type" means a category of power-driven vehicles which do not differ in such essential respects as:

2.2.1. the manufacturer's indications of the vehicle type,

2.2.2. the arrangement and design of the vehicle component or components on which the protective device acts,

2.2.3. the type of protective device,

2.3. "Protective device" means a system designed to prevent unauthorized use of the vehicle providing a positive locking action of the steering or of the transmission; this system may be:

2.3.1. type 1: solely and positively operated on the steering alone,

2.3.2. type 2: positively operated on the steering in conjunction with the device which de-activates the engine of the vehicle,

2.3.3. type 3: pre-loaded, operating on the steering in conjunction with the device which de-activates the engine of the vehicle;

2.3.4. Type 4: positively operated on the transmission.

2.4. "Steering" means the steering control (handlebars), the steering head and its accessory cladding, the steering shaft and all other components which directly affect the effectiveness of the protective device;

2.5. "Combination" means one of the specifically planned and constructed variations of a locking system which, when properly activated, permits operation of the locking system;

2.6. "*Key*" means any mechanical and/or electronic solution designed and constructed to provide a method of operating a locking system which is designed and constructed to be operated by that mechanical and/or electronic solution.

2.7. "*Primary user*" is a user who is able to authorize digital keys. There can be more than one primary users.

2.8. "*Digital key*" means a key designed to be transferred to multiple devices by the primary user(s) through dedicated processes.

2.9. "*Close proximity*" means a distance of less than 6 m.

3. APPLICATION FOR APPROVAL

3.1. The application for approval of a vehicle type with regard to a protective device to prevent its unauthorized use shall be submitted by the vehicle manufacturer or by his duly accredited representative.

3.2. It shall be accompanied by the undermentioned documents in triplicate and by the following particulars:

3.2.1. a detailed description of the vehicle type with regard to the arrangement, and design of the vehicle component or components on which the protective device acts;

3.2.2. drawings, on an appropriate scale and in sufficient detail, of the protective device and of its mountings on the vehicle;

3.2.3. a technical description of the device.

3.3. There shall be submitted to the technical service responsible for conducting the approval tests:

3.3.1. a vehicle, representative of the vehicle type to be approved, if requested by the technical service; and also

3.3.2. at the request of the technical service, such components' of the vehicle as the service deems essential for the checks prescribed in paragraphs 5. and 6. of this Regulation.

4. APPROVAL

4.1. If the vehicle submitted for approval pursuant to this Regulation meets the requirements of paragraphs 5. and 6. below, approval of that vehicle type shall be granted.

4.2. An approval number shall be assigned to each type approved. Its first two digits (at present 00 for the Regulation in its original form) shall indicate the series of amendments incorporating the most recent major technical amendments made to the Regulation at the time of issue of the approval. The same Contracting Party may not assign the same number either to the same vehicle type equipped with another type of protective device or whose protective device is mounted differently, or to another vehicle type.

4.3. Notice of approval or of refusal of approval of a vehicle type pursuant to this Regulation shall be communicated to the Parties to the Agreement which apply this Regulation, by means of a form conforming to the model in annex 1 to this Regulation and of drawings of the protective device and its mounting supplied by the applicant for approval, in a format not exceeding A4 (210 x 297 mm) or folded to that format and on an appropriate scale.

4.4. There shall be affixed, conspicuously and in a readily accessible place specified on the approval form, to every vehicle conforming to a vehicle type approved under this Regulation an international approval mark consisting of:

4.4.1. a circle surrounding the letter "E" followed by the distinguishing number of the country which has granted approval;[2/]

2/ 1 for Germany, 2 for France, 3 for Italy, 4 for the Netherlands, 5 for Sweden, 6 for Belgium, 7 for Hungary, 8 for the Czech Republic, 9 for Spain, 10 for Serbia and Montenegro, 11 for the United Kingdom, 12 for Austria, 13 for Luxembourg, 14 for Switzerland, 15 (vacant), 16 for Norway, 17 for Finland, 18 for Denmark, 19 for Romania, 20 for Poland, 21 for Portugal, 22 for the Russian Federation, 23 for Greece, 24 for Ireland, 25 for Croatia, 26 for Slovenia, 27 for Slovakia, 28 for Belarus, 29 for Estonia, 30 (vacant), 31 for Bosnia and Herzegovina, 32 for Latvia, 33 (vacant), 34 for Bulgaria, 35 (vacant), 36 for Lithuania, 37 for Turkey, 38 (vacant), 39 for Azerbaijan, 40 for The former Yugoslav Republic of Macedonia, 41 (vacant), 42 for the European Community (approvals are granted by its Member States using their respective ECE symbol), 43 for Japan, 44 (vacant), 45 for Australia, 46 for Ukraine, 47 for South Africa, 48 for New Zealand, 49 for Cyprus, 50 for Malta and 51 for the Republic of Korea. Subsequent numbers shall be assigned to other countries in the chronological order in which they ratify or accede to the Agreement Concerning the Adoption of Uniform Technical Prescriptions for Wheeled Vehicles, Equipment and Parts which can be Fitted and/or be Used on Wheeled Vehicles and the Conditions for Reciprocal Recognition of Approvals Granted on the Basis of these Prescriptions, and the numbers thus assigned shall be communicated by the Secretary-General of the United Nations to the Contracting Parties to the Agreement.

4.4.2. the number of this Regulation, followed by the letter "R'", a dash and the approval number to the right of the circle described in paragraph 4.4.1.

4.5. If the vehicle conforms to a vehicle type approved under one or more other Regulations annexed to this Agreement in the country which has granted approval under this Regulation, the symbol prescribed in paragraph

4.4.1 need not be repeated; in such case the regulation and approval numbers and the additional symbols of all the Regulations under which approval has been granted in the country which has granted approval under this Regulation shall be placed in vertical columns to the right of the symbol prescribed in paragraph 4.4.1.

4.6. The approval mark shall be clearly legible and be indelible.

4.7. The approval mark shall be placed close to or on the vehicle data plate affixed by the manufacturer.

4.8. Annex 2 to this Regulation gives examples of arrangements of approval marks.

5. GENERAL SPECIFICATIONS

5.1. The protective device shall be so designed that:

5.1.1. it is necessary to put it out of action in order to enable the vehicle to be steered, or to be driven or moved forward in a straight line,

5.1.2. In the case of protective devices of type 4, the device shall be so designed that it is necessary to put it out of action in order to release the transmission. If this device is activated by the control of the parking device it must act in conjuction with the device which deactivates the engine of the vehicle.

5.1.3. it shall only be possible to extract the key with the bolt in the fully engaged or in the fully disengaged position. Any intermediate position of the key risks subsequent engagement of the bolt, even if the key of the protective device is inserted, shall be excluded.

5.2. The requirements of paragraph 5.1. shall be met by the single application of one key.

5.3. The protective device referred to in paragraph 5.1 above, and the vehicle components on which it operates, shall be so designed, that it cannot rapidly and without attracting attention, be opened, rendered ineffective, or destroyed by, for example, the use of low-cost, easily-concealed tools, equipment or fabrications readily available to the public at large.

5.4. The protective device shall be mounted on the vehicle as an item of original equipment (i.e. equipment installed by the vehicle manufacturer prior to first retail sale). The lock shall be securely assembled in the protective device. (If the lock can be extracted using the key and after the cover or any other retention device has been removed,. this is not in contradiction with the requirement).

5.5. The key looking system shall provide at least 1,000 different key combinations or a number equal to the total number of vehicles manufactured annually if less than 1,000. In vehicles of one type the frequency of occurrence of each combination shall be roughly one per 1,000.

5.6. The key and lock shall not be visibly coded.

5.7. The look shall be so designed, constructed and fitted that turning of the lock cylinder, when in the locked position, with a torque of less than 0,245 mdaN is not possible with anything other than the mating key, and

5.7.1. for lock cylinders with pin tumblers no more than two identical tumblers operating in the same direction shall be positioned adjacent to each other, and in a lock there shall not be more than 60 per cent identical tumblers,

5.7.2. for lock cylinders with disc tumblers no more than two identical tumblers operating in the same direction shall be positioned adjacent to each other, and in a lock there shall not be more than 50 per cent identical tumblers.

5.8. Protective devices shall be such as to exclude any risk, while the vehicle is in motion with engine running, of accidental blockage likely to compromise safety in particular.

5.9. The protective device, if it is of type 1, type 2 or type 3, shall, in its activated position, be strong enough to withstand, without damage to the steering mechanism likely to compromise safety, the applicatiom of a torque of 20 mdaN about the axis of the steering shaft in both directions under static conditions.

5.10. The protective device, if it is of type 1, type 2 or type 3, shall be so designed that the steering can only be locked at an angle of at least 20° to the left and/or the right of the straight ahead position.

5.11. Electromechanical and electronic devices to prevent unauthorized use

Electromechanical and electronic devices to prevent unauthorized use, where fitted, shall comply with the requirements of paragraphs 5. and 6., mutatis mutandis.

Components that are not embedded in the vehicle (e.g. keys, which are used for activation/deactivation) need not to comply with the test requirements described in paragraphs 5. and 6.

If the technology of the device is such that paragraphs 5. and 6. are not applicable, it shall be verified that care has been taken to preserve safety of the vehicle. The functioning process of these devices shall incorporate secure means to prevent any risk of blocking or accidental dysfunction that could compromise the safety of the vehicle.

5.12. Electromagnetic compatibility

Electromechanical and electronic devices to prevent unauthorized use, where fitted, shall comply with the following requirements regarding electromagnetic compatibility.

Tests shall be performed according to the relevant technical prescriptions and transitional provisions of UN Regulation No. 10, 06 series of amendments and according to the immunity test methods described in Annex 6 and according to the emission test methods described in Annexes 4 and 5.

5.13. In addition, digital keys shall comply with the provisions of Annex 4.

6. PARTICULAR SPECIFICATIONS

6.1. In addition to the general specifications prescribed in paragraph 5, the protective device shall comply with the particular conditions prescribed below:

6.1.1. In the case of protective devices of type 1 or type 2, it shall only be possible to engage the lock by means of a movement of the key, the handlebars being in the position appropriate for the engagement of the bolt in the corresponding slot.

6.1.2. In the case of protective devices of type 3, it shall only be possible to pre-load the bolt by a separate action on the part of the user of the vehicle, combined with or in addition to the rotation of the key. It shall not be possible to remove the key once the bolt has been pre-loaded, except in accordance with the provisions of paragraph 5.1.3. above.

6.2. In the case of protective devices of type 2 and type 3, it shall not be possible for the bolt to engage so long as the device is set in a position which permits the activation of the engine of the vehicle.

6.3. In the case of protective devices of type 3, when the device is set to act, it shall not be possible to prevent the device from functioning.

6.4. In the case of protective devices of type 3, the protective device must remain in good working order and must, in particular, continue to meet the requirements of paragraphs 5.7., 5.8., 5.9. and 6.3 above after it has undergone 2,500 locking cycles in each direction of the test specified in annex 3 of this Regulation.

7. MODIFICATIONS OF VEHICLE TYPE OR OF THE VEHICLE'S PROTECTIVE DEVICE

7.1. Every modification of the vehicle type or of the vehicle's protective device shall be notified to the administrative department which approved the vehicle type. The department may then either:

7.1.1. consider that the modifications made are unlikely to have an appreciable adverse .effect, and that in any case the vehicle still complies with the requirements; or

7.1.2. require a further test report from the technical service responsible for conducting the tests.

7.2. Confirmation or refusal of approval, specifying the alterations, shall be communicated by the procedure specified in paragraph 4.3. above to the Parties to the Agreement which apply this Regulation.

8. CONFORMITY OF PRODUCTION

8.1. Every vehicle bearing an approval mark as prescribed under this Regulation shall conform, with regard to the type of protective device, to the mounting of the latter on the vehicle, and to the components on which the protective device acts, to the vehicle type approved.

8.2. In order to verify conformity as prescribed in paragraph 8.1. above a sufficient number of random checks shall be made on serially-manufactured vehicles bearing the approval mark required by this Regulation.

9. PENALTIES FOR NON-CONFORMITY OF PRODUCTION

9.1. The approval granted in respect of a vehicle type pursuant to this Regulation may be withdrawn if the requirements laid down in paragraph 8.1. above are not complied with.

9.2. If a Party to the Agreement which applies this Regulation withdraws an approval it has previously granted, it shall forthwith so notify the other Contracting Parties applying this Regulation, by means of a copy of the approval form beaming at the end, in large letters, the signed and dated annotation "APPROVAL WITHDRAWN".

10. PRODUCTION DEFINITELY DISCONTINUED

If the holder of the approval completely ceases to manufacture a type of vehicle approved in accordance with this Regulation, he shall so inform the authority which granted the approval. Upon receiving the relevant communication that authority shall inform thereof the other Parties to the Agreement applying this Regulation by means of a copy of the approval form bearing at the end, in large letters, the signed and dated annotation: "PRODUCTION DISCONTINUED".

11. NAMES AND ADDRESSES OF TECHNICAL SERVICES RESPONSIBLE FOR CONDUCTING APPROVAL TESTS, AND OF ADMINISTRATIVE DEPARTMENTS

The Parties to the Agreement applying this Regulation shall communicate to the United Nations Secretarial the names and addresses of the technical services responsible for conducting approval tests and of the administrative departments which grant approval and to which forms certifying approval or refusal or withdrawal of approval, issued in other countries, are to be sent.

12. Transitional provisions

12.1. As from the official date of entry into force of the 01 series of amendments, no Contracting Party applying this Regulation shall refuse to grant or refuse to accept type approvals under this Regulation as amended by the 01 series of amendments.

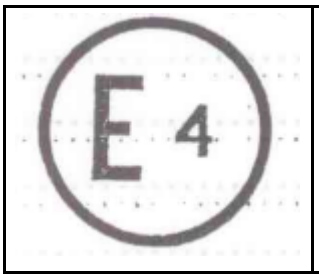
12.2. As from 1 September 2022, Contracting Parties applying this Regulation shall not be obliged to accept type approvals to the preceding series of amendments, first issued after 1 September 2022.

12.3. Until 1 September 2024, Contracting Parties applying this Regulation shall accept type approvals to the preceding series of amendments, first issued before 1 September 2022.

12.4. As from 1 September 2024, Contracting Parties applying this Regulation shall not be obliged to accept type approvals issued to the preceding series of amendments to this Regulation.

Annex 1 (to the Regulation)

(Maximum format: A4 (210 × 297 mm))



Name of administration

Communication .concerning the approval (or refusal or withdrawal of approval or production definitely discontinued) of a type of power-driven vehicle with handlebars with regard to its protection against unauthorized use, pursuant to Regulation No.62

Approval No. .....

1. Trade name or mark of the power-driven vehicle .....

2. Vehicle type .....

3. Manufacturer's name and address .....

4. If applicable, name and address of manufacturer's representative .....

5. Brief description of the protective device, of its mounting, and function and of the steering system of the vehicle. .....

6. Vehicle submitted for approval on .....

7. Technical service responsible for conducting approval tests .....

8. Date of report issued by that service .....

9. Number of report issued by that service .....

10. Approval granted/refused 1/ .....

1/ Strike out what does not apply.

11. Position of approval mark on the vehicle .....

12. Place .....

13. Date .....

14. Signature .....

15. The following documents, bearing the approval number shown above, are annexed to this communication:

..... drawings, diagrams and plans of the protective device, of its mounting, and of the vehicle components on which it acts;

..... photographs of the protective device.

Annex 2 ARRANGEMENTS OF APPROVAL MARKS

Model A

(See paragraphs 4.1.4. and 4.2.4. of this Regulation)

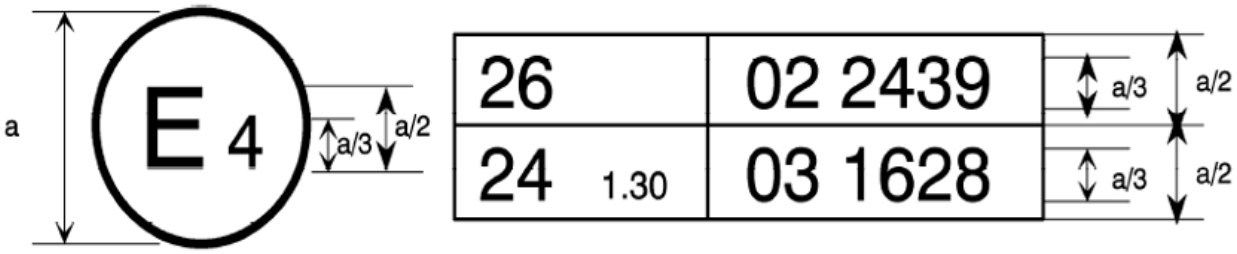


	a= 8 mm min

The above approval mark affixed to a vehicle shows that the vehicle type concerned has been approved in the Netherlands (E 4) pursuant to UN Regulation No. 62 and under approval number 012439. The first two digits of the approval number indicate that the approval was granted in accordance with the requirements of UN Regulation No. 62 as amended by the 01 series of amendments.

Model B

(See paragraph 4.1.5. of this Regulation)



	a= 8 mm min

The above approval mark affixed to a vehicle shows that the vehicle type concerned has been approved in the Netherlands (E 4) pursuant to UN Regulation No. 62and UN Regulation No. 10. The first two digits of the approval numbers indicate that, at the dates when the respective approvals were given, UN Regulation No. 62 included the 01 series of amendments, while UN Regulation No. 10 already included the 06 series of amendments.

1/ The second number is given merely as an example.

Annex 3 TEST FOR WEAR IN TYPE 3 PROTECTIVE DEVICES

1. **TEST EQUIPMENT**

1.1. The test equipment shall consist of:

1.1.1. a fixture suitable for mounting the sample steering complete with the protective device attached, as defined in paragraph 2.3. of this Regulation;

1.1.2. a means for activating and de-activating the protective device which shall include the use of the key;

1.1.3. a means for rotating the steering shaft relative to the protective device.

2. **TEST METHOD**

2.1. A sample of the steering complete with the protective device is attached to the fixture referred to in paragraph 1.1.1. above.

2.2. One cycle of the test procedure shall consist of the following operations:

2.2.1. **Start position**

The protective device shall be de-activated and the steering shaft shall be rotated to a position which prevents engagement of the protective device.

2.2.2. **Set to activate**

The protective device shall be moved from the de-activated to the activated position, using the key.

2.2.3. **Activated**

The steering shaft shall be rotated such that the torque on it, at the instant of engagement of the protective device shall be 5.88 Nm ± 0.25.

2.2.4. **De-activated**

The protective device shall be de-activated by the normal means, the torque being reduced to zero to facilitate disengagement.

2.2.5. **Return**

The steering shaft shall be rotated to a position which prevents engagement of the protective device.

2.2.6. **Opposite rotation**

Repeat procedures described in paragraphs 2.2.2., 2.2.3., 2.2.4., and 2.2.5. but in the opposite direction of rotation of the steering shaft.

2.2.7. The time interval between two successive engagements of the device shall be at least 10 seconds.

2.3. The wear-producing cycle shall be repeated for the number of times specified in paragraph 6.4. of this Regulation.

Annex 4 Safety Provisions for Digital Keys

1. General

The purpose of this Annex is to specify the requirements for documentation and verification for digital keys used to operate the ‘device to prevent unauthorized use’ of the vehicle.

2. Definitions

2.1. "*Authorization process*" means any method to provide the digital key which can operate the ‘device to prevent unauthorized use’ of the vehicle.

2.2. "*Revocation process*" means any method to prevent the digital key to operate the ‘device to prevent unauthorized use’ of the vehicle.

2.3. "*Boundary of functional operation*" defines the boundaries of the external physical limits (e.g. distance) within which the digital key is able to operate the ‘device to prevent unauthorized use’ of the vehicle.

3. Documentation

The vehicle manufacturer shall provide the following documentation for type approval:

3.1. A description of the authorization process.

3.2. A description of the revocation process.

3.3. A description of the boundary of functional operation.

3.4. A description of the safety measures designed within the digital key revocation process to ensure safe operation of the vehicle.

4. Requirements for Safe Operation

4.1. A digital key shall only be transferred to a device via the authorization process.

4.2. There shall be a revocation process.

4.2.1. Revocation of a digital key shall not result in an unsafe condition.

A risk reduction analysis using functional safety standard such as ISO 26262 and safety of the intended functionality standard such as ISO 21448, which documents the risk to vehicle occupants caused by revocation of a digital key and documents the reduction of risk resulting from implementation of the identified risk mitigation functions or characteristics, shall be performed.

4.2.2. It shall be possible for the primary user(s) to identify the number of authorized registered digital keys.

4.3. Boundary of functional operation for the device to prevent unauthorized use:

4.3.1. Unlocking of the device to prevent unauthorized use shall require that an authorized registered digital key is detected in the interior of the vehicle, or in close proximity of the vehicle.

4.4. Detailed information shall be contained in the owner's manual of the vehicle, or by any other communication means in the vehicle; as a minimum, this information shall include:

(a) The method(s) for authorization of the digital key;

(b) The method(s) for revocation of the digital key.

5. The effectiveness of the system shall not be adversely affected by cyber-attacks, cyber threats and vulnerabilities. The effectiveness of the security measures shall be demonstrated by fulfilling the technical requirements of UN Regulation No. 155.

6. Verification

Verification of the functionality of the digital key shall be conducted with the support of the manufacturer's documentation as specified in paragraph 3.

7. Competence of the Auditors and the Assessors

The assessments under this Annex shall only be conducted by auditors or assessors with the technical and administrative knowledge necessary for such purposes. They shall, in particular, be competent as auditor/assessor for ISO 26262-2018 (Functional Safety – Road Vehicles), and ISO 21448 (Safety of the Intended Functionality – Road Vehicles); and shall be able to make the necessary link with cybersecurity aspects in accordance with UN Regulation No. 155 and ISO/SAE 21434. This competence should be demonstrated by appropriate qualifications or other equivalent training records.